

# Responsible AI

ECE 750T

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Department of Electrical and Computer Engineering

University of Waterloo

# 0 Course information

# Territorial acknowledgement



We acknowledge that we live and work on the traditional territory of the **Neutral, Anishinaabeg, and Haudenosaunee** peoples.

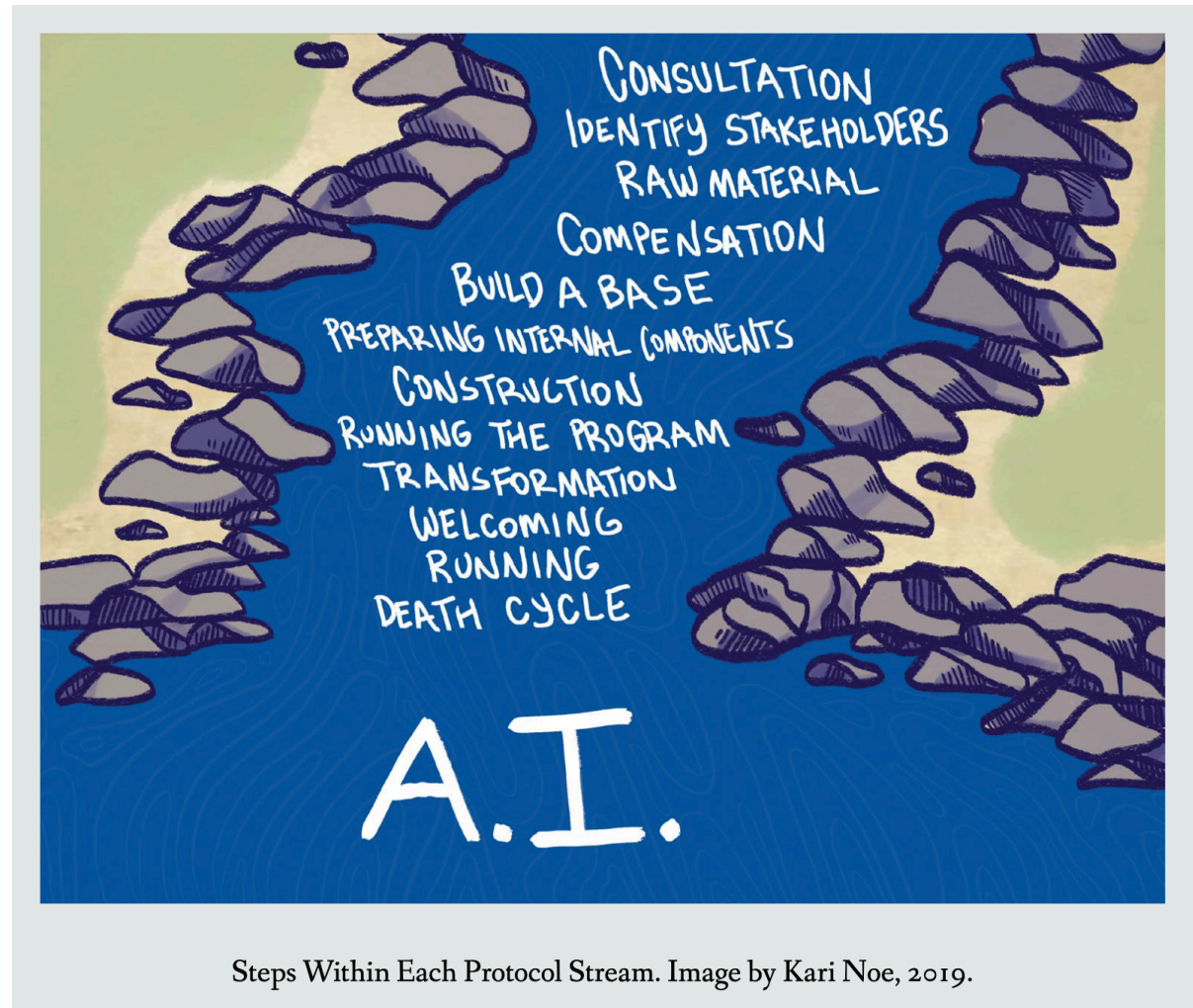
The University of Waterloo is situated on the **Haldimand Tract**, the land promised to the Six Nations that includes six miles on each side of the Grand River.

# Indigenous Protocol and Artificial Intelligence

## How to Build Anything Ethically

Suzanne Kite et al.

Indigenous Protocol and  
Artificial Intelligence Position  
Paper (2020)



# whoami



**Elliot Creager**  
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**Hobbies:** Guitar 🎸, birdwatching 🐦

## Education: 🎓

- Ph.D., Computer Science  
@ University of Toronto
- M.A., Music Technology  
@ McGill University
- Sc.B., Electrical Engineering; A.B., Music  
@ Brown University

## Employment: 💻

- Research Intern  
@ Google Brain
- Research Scientist  
@ Analog Devices

# whoarewe — Ambreesh



**Ambreesh Parthasarathy**  
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**Hobbies:** Music production 🎵, football ⚽,  
photography 📷

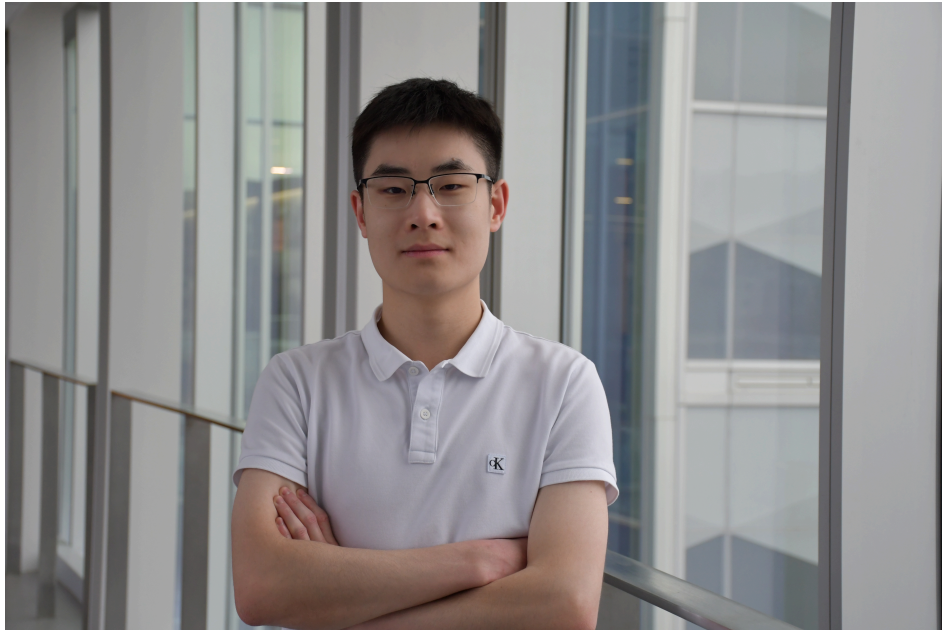
## Education: 🎓

- M.A.Sc., Electrical & Computer Engineering  
@ University of Waterloo
- B.Tech., Computer Science  
@ Shiv Nadar University

## Employment: 💻

- Pre-doctoral Fellow  
@ Centre for Responsible AI, IITM
- Analyst (Intern)  
@ McKinsey & Co

# whoarewe — Chenghao



**Chenghao Tan**  
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## Education: 🎓

- M.A.Sc., Electrical & Computer Engineering  
@ University of Waterloo
- B.Eng., Computer Science  
@ Hangzhou Dianzi University

## Hobbies:

- Movies 🎬
- Traveling 🌍
- Open-source DIY 🔧

# Who are you?

- Graduate students with a foundation in machine learning
- Different research interests and experiences
- A shared interest in building AI responsibly

What program are you in, and what experience do you bring? 🤔

Answer in class.

# Where are we?

| Meeting                 | Time                 | Location |
|-------------------------|----------------------|----------|
| Lecture                 | Fridays, 1:00–3:50pm | PSE 4053 |
| Instructor office hours | By appointment       | EIT 3116 |

- **LEARN:** announcements and course materials.
- **Piazza:** course questions and finding a presentation group; link forthcoming on LEARN.
- **Email the instructor:** personal matters, accommodations, illness, or missed work.
- **Lectures will not be recorded.**

# What will we learn?

## Understand the risks

- Fairness and measurement
- Evaluation and auditing
- Memorization and privacy
- Foundation-model safety

## Assess possible responses

- Fair learning and private learning
- Data attribution and unlearning
- Data poisoning and collective action
- Governance and policy

Read research critically, lead a discussion, and carry out an **individual research project**.

# What would you like to learn?

What does “Responsible AI” mean to you? 🤔

- Answer in class.

Which AI risks or questions would you like us to address? 🤔

- Answer in class.

hallucination  
evaluation  
monitoring  
proactive goals

control  
structural risk  
x risk  
governance  
regulation

accountability  
privacy preservation  
data provenance

# How class will work

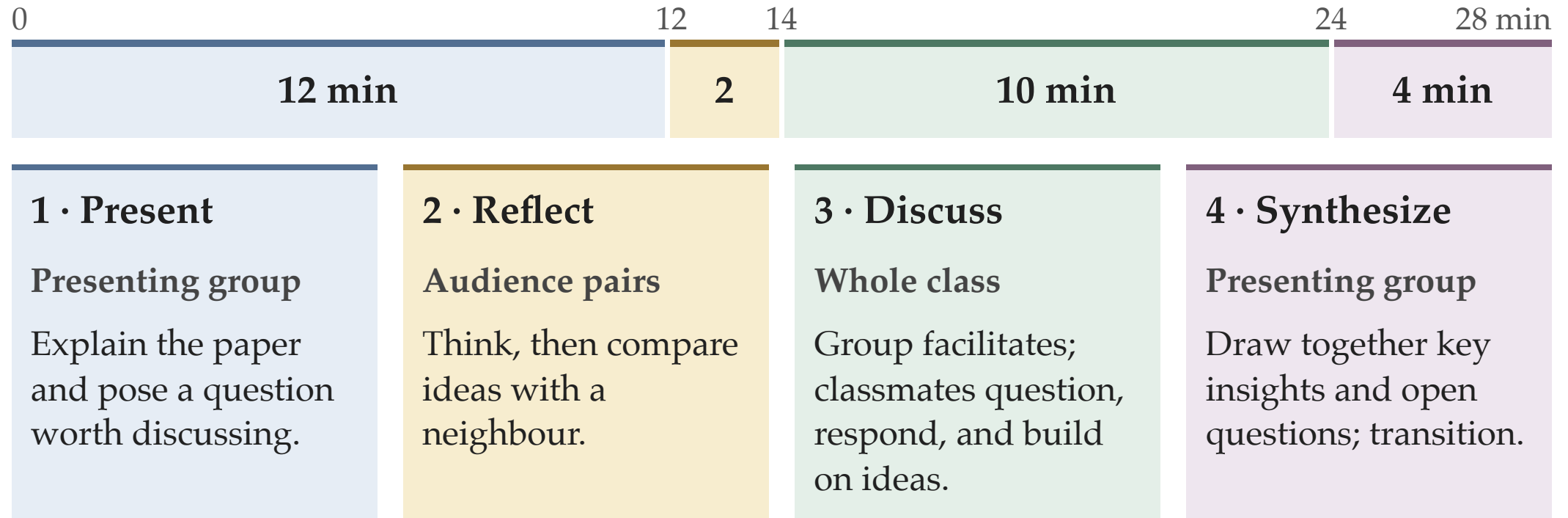
- **Weeks 1–2:** prof lectures and class discussion.
- **Weeks 3–10:** prof lecture, break, three student-led paper sessions.
  - student presentations in **groups of three**. sign-up sheet TBA.
  - you will present **once** this term. think about which week/topic you want to explore!
- **Week 6:** midterm exam replaces prof lecture.
- **Week 12:** individual research-project presentations.
- **Week ??:** guest lectures?

# Assessment at a glance

| Component                            | Weight | Work submitted by                |
|--------------------------------------|--------|----------------------------------|
| Paper presentation & discussion      | 15%    | Group (3 students) — shared mark |
| Micro-survey                         | 5%     | Group (3 students) — shared mark |
| Assignment 1: Fairness               | 10%    | Each student                     |
| Assignment 2:<br>Documentation audit | 10%    | Each student                     |
| Midterm exam                         | 10%    | Each student                     |
| Research project                     | 50%    | Each student                     |

**Project breakdown:** proposal 5% · report 35% · presentation 10%.

# Group presentation outline



**Your group leads → your group scribes.** Divide responsibilities within the group.

Focus notes on **discussion and synthesis**; your presentation slides will also be shared.

Send the professor **Markdown notes** (write in Markdown or convert before sending). Notes are shared with the class.

# Lead a discussion worth having

**Shared 15% mark: 10% presentation + 5% facilitation.**

- Explain the important ideas and evidence clearly.
- Ask accessible questions and follow up on classmates' reasoning.
- Connect contributions and synthesize what the discussion revealed.
- **Your group also scribes.** Capture the discussion; slides and notes will be shared with the class.

# Post-discussion: writing a micro-survey

**5%, shared group mark · Due Nov 27**

- Compare your presented paper with **two others from the same week's reading list**.
- Connect their questions, assumptions, approaches, and findings.
- Incorporate an insight or question from the class discussion.
- Include a brief **author contribution statement**.

# Assignments and midterm

| Task                          | Out    | Due / exam       |
|-------------------------------|--------|------------------|
| A1: Fairness — 10%            | Sep 18 | Oct 2            |
| A2: Documentation audit — 10% | Oct 3  | Oct 19           |
| Midterm exam — 10%            | —      | Oct 23, in class |

- **A1:** fair classification, metrics, an intervention, and reflection.
- **A2:** audit dataset / model documentation and the validity of a measurement; no programming.
- **Midterm:** Weeks 1–5; conceptual questions and worked problems.

# Your individual research project

**50% of the course grade. All components are individual.**

| Component              | Weight | Deliverable                           |
|------------------------|--------|---------------------------------------|
| Proposal — Week 5      | 5%     | 1–2 pages: question, motivation, plan |
| Report — end of term   | 35%    | 6–8 pages, excluding references       |
| Presentation — Week 12 | 10%    | Explain your research and findings    |

Choose a research question connected to the course and develop an appropriate methodology.

# Submissions and deadlines

- Unless otherwise instructed, submit written work to **both Crowdmark and LEARN Dropbox**.
- Written submissions are due at **11:59pm** on the stated date.
- **Late submissions will not be accepted**, except where an accommodation applies.
- Micro-surveys: one submission per group.

See the course outline and assignment instructions for details.

# Course policies: generative AI

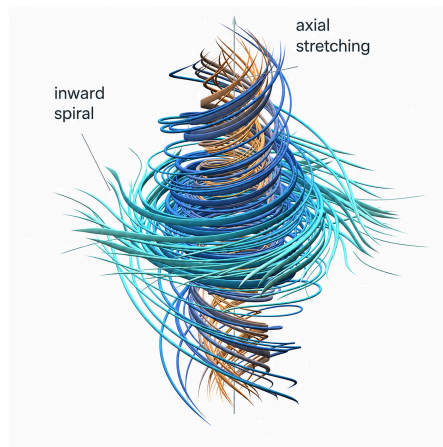
Generative AI tools produce text, images, music, or code—for example, ChatGPT, DALL-E, or GitHub Copilot. See the course outline for the full policy. 📌

- **GenAI may be used with proper documentation, citation, and acknowledgment ([how to cite](#)).**
- GenAI may falsify references, fabricate facts, or inaccurately express ideas.
- Content generated from other authors' work may contain inaccuracies or reflect biases.
- The legal and copyright status of GenAI inputs and outputs is unclear; see the [Copyright Advisory Committee](#).
- **You are accountable for the content and accuracy of all submitted work, including work supported by GenAI.**
- **Clarity matters:** Submitting “sloppy” work may result in poor marks.
  - Be prepared to reproduce any part of their submitted work for the instructor upon request

# 1 AI Risks and Taxonomies

# What's New?

## The Navier–Stokes controversy: AI, credit, and trust



For most of the past year progress was slow. We worked through the literature and upgraded various preliminary results, up to obtaining finite time blow up for the Incompressible Porous Media equation (with smooth forcing). This was until about a month ago, when we had real progress: on August 15th, we obtained the blow up results, with smooth forcing, for both Boussinesq and Euler. I can say the first LLM generated proof Levent sent me was the most horrendous I have ever read; we verified it on Lean on August 22nd. Since this point, we have been working around the clock to understand this proof and turn it into something readable.

There is another part of this story, and one that, honestly, I very much wish I did not have to be concerned with. I am not happy about the presentation quality in these papers. Ideally, we would have preferred to spend weeks turning the LLM generated proofs into something readable from the very first page.

## Verification · Readability · Credit

How should researchers disclose AI use, and how should credit for new discoveries be assigned? 🤔

11 Sept. 2026 · Visualization: OpenAI (blog). Annotated statement: Buckmaster (2026), p. 1.

# What are we asking AI to do?

## Perceive

Identify an object in an image.

## Predict

Estimate an outcome from records.

## Generate

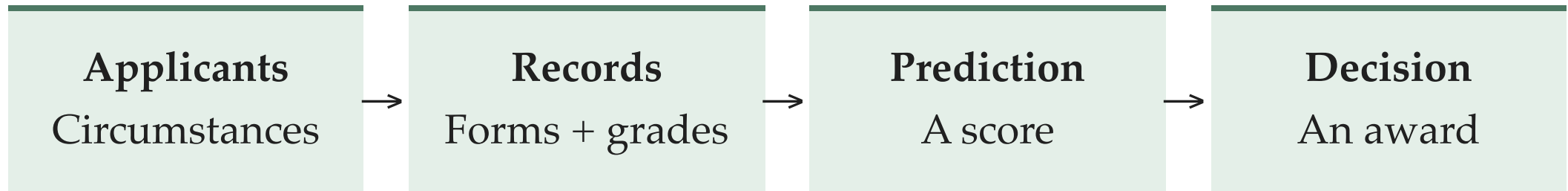
Draft text, images, or code.

## Control

Choose actions as a situation unfolds.

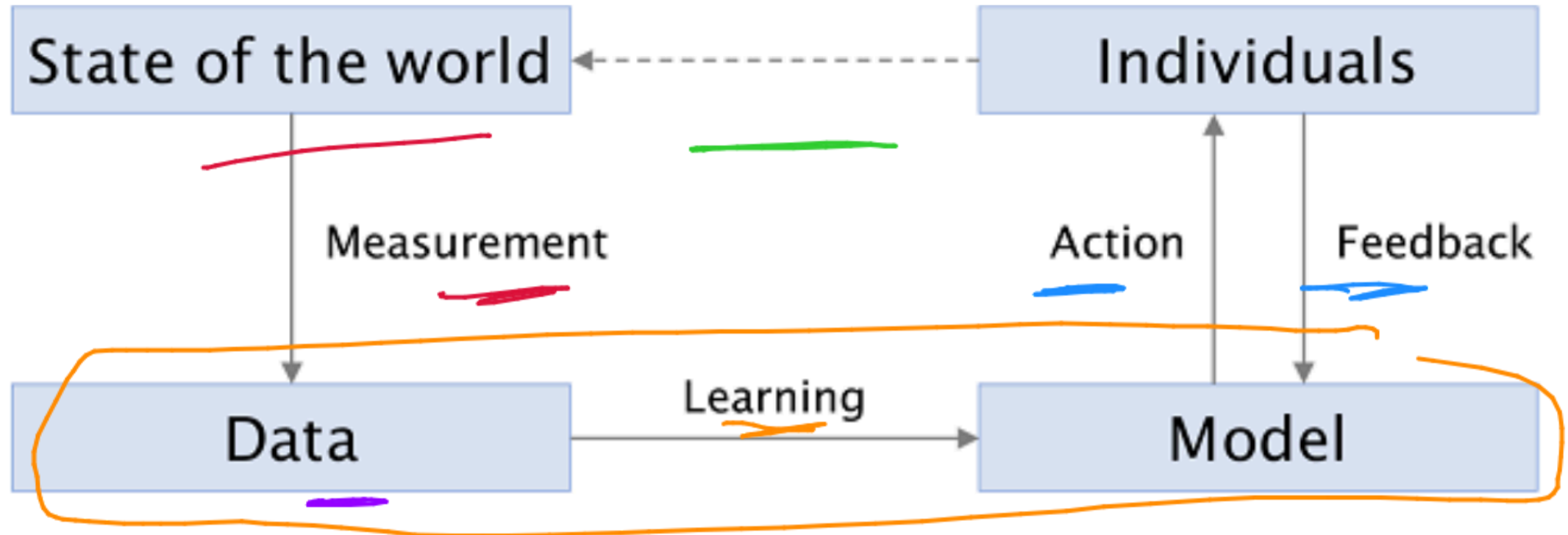
# A scholarship service

**Our hypothetical case:** a university has a limited scholarship budget.



What should a scholarship help us achieve? 🤔

# The machine learning loop



**The model sits inside a larger system.**

Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 1, p. 5.

# What counts as “success”?

**Goal:** help students flourish

**Future grades?**

Academic performance

**Financial need?**

Access to resources

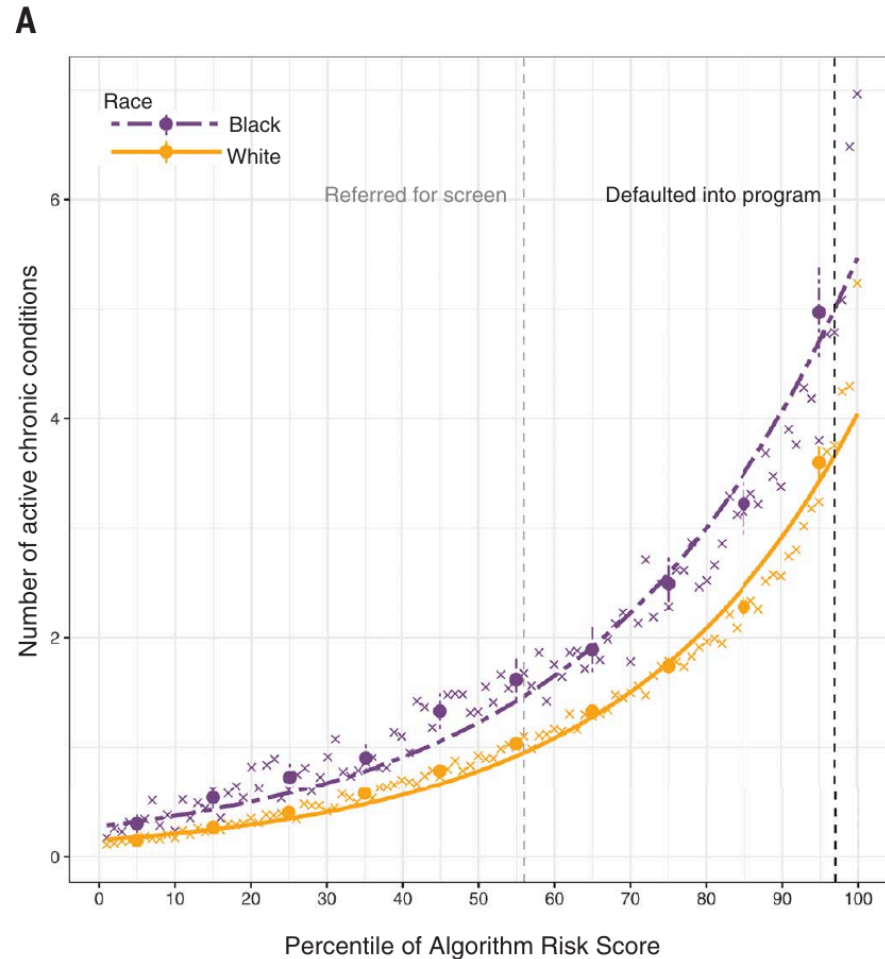
**Past awards?**

Previous decisions

What would each target leave out? 🤔

Hypothetical application of Barocas, Hardt, and Narayanan (2023), Chapter 1, “Measurement,” pp. 6–9.

# When cost stands in for need



## Intended use

Identify patients needing extra care.

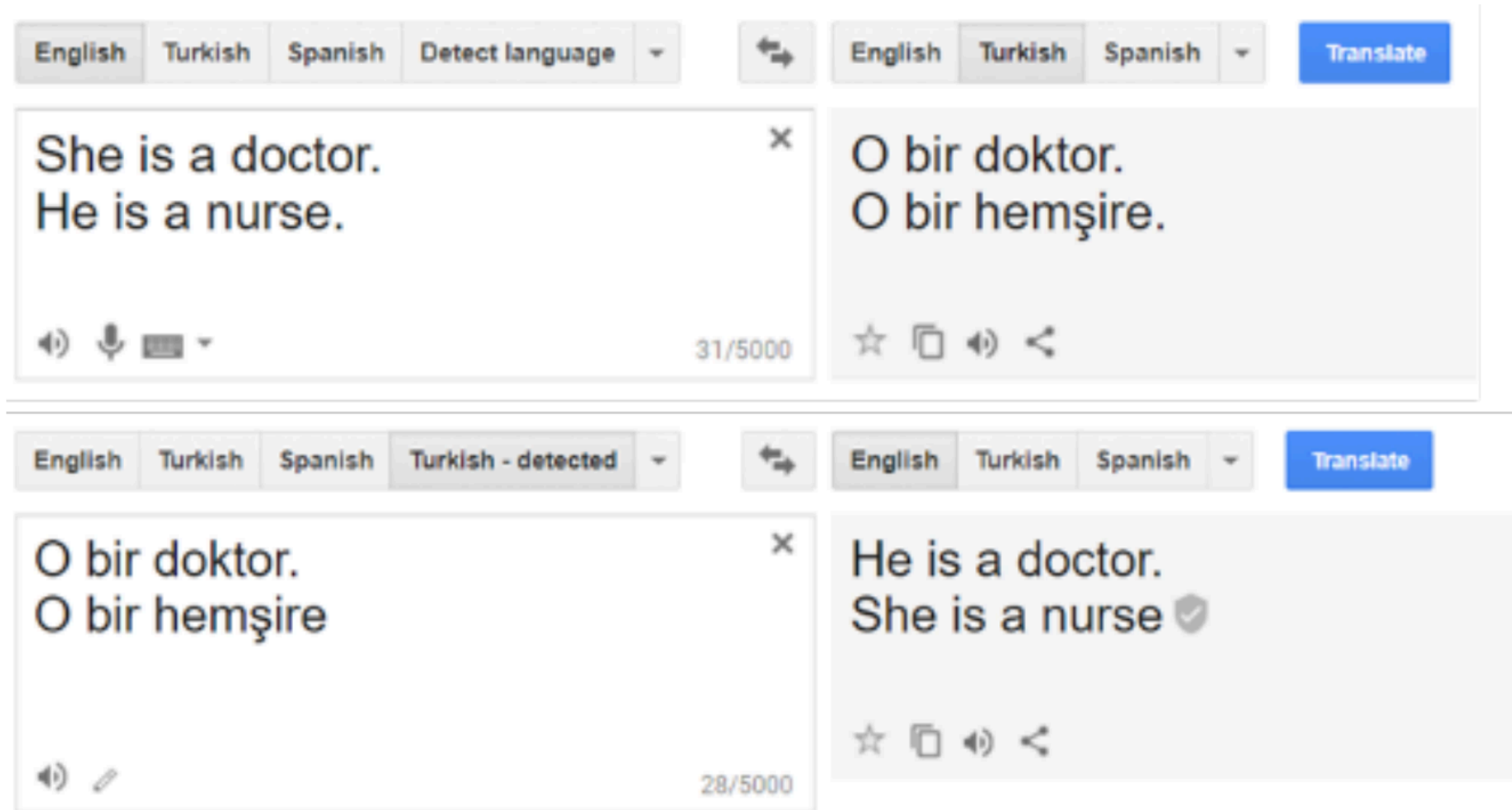
## Prediction target

Future health-care costs.

At the **same score**, Black patients had more chronic illnesses.

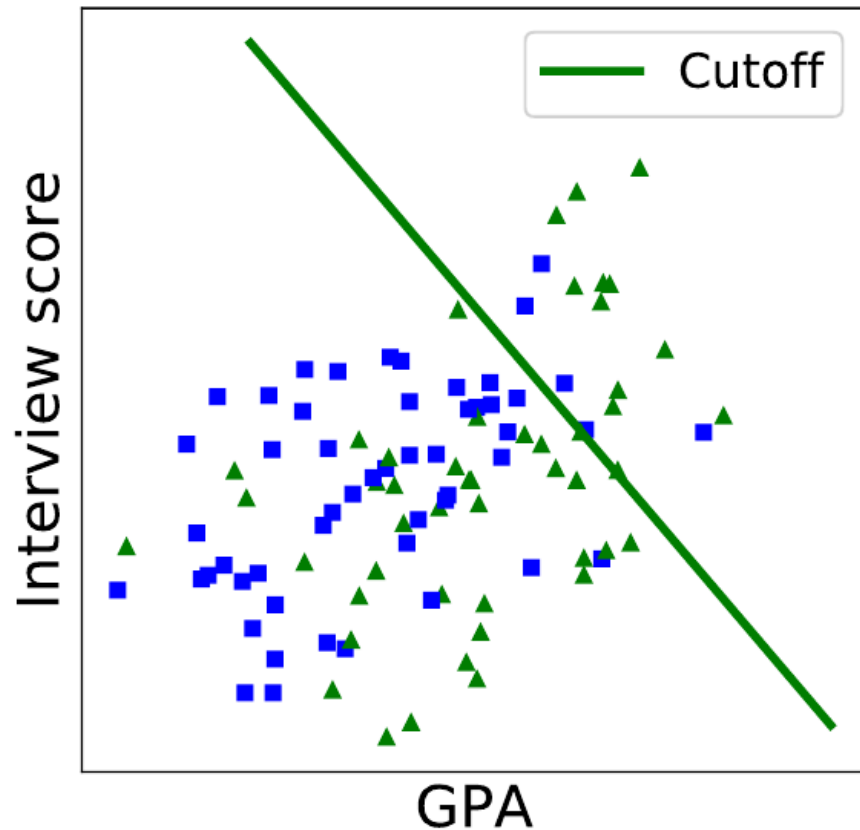
Obermeyer et al. (2019), Fig. 1A. One studied health-care algorithm.

# Which patterns should a model reproduce?



Historical example reproduced in Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 3, p. 11.

# A toy hiring example



Each marker is an applicant.

■ Group A · ▲ Group B

Groups represent **demographic categories**, such as gender or race.

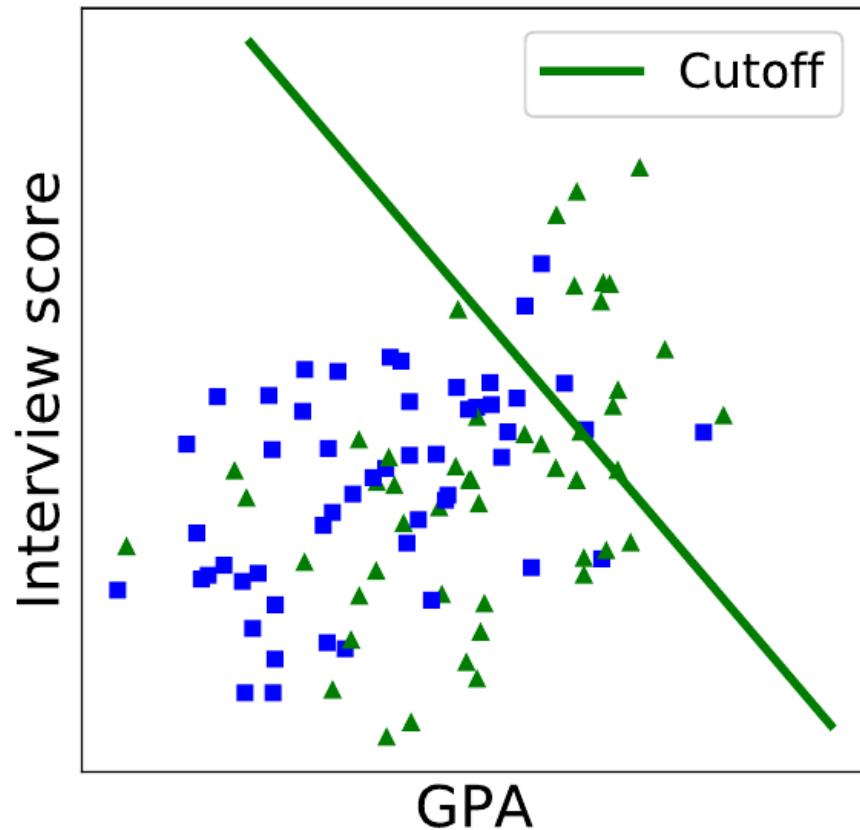
The toy example assigns **no specific identities** to A or B.

**Position:** GPA + interview score

**Line:** hiring cutoff

Synthetic illustration. Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 4, pp. 15–16.

# Is removing the group label enough?



**Inputs:** GPA + interview score

**Target:** later job-performance rating

**Same hiring rule** for A and B.

Group membership is **not** an input.

Is that enough to establish fairness? 🤔

Toy hiring example—not scholarship data. Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 4, pp. 15–17.

# A prediction does not choose a policy

**One prediction:** expected future grade

## **Rank**

Fund the highest scores.

## **Combine**

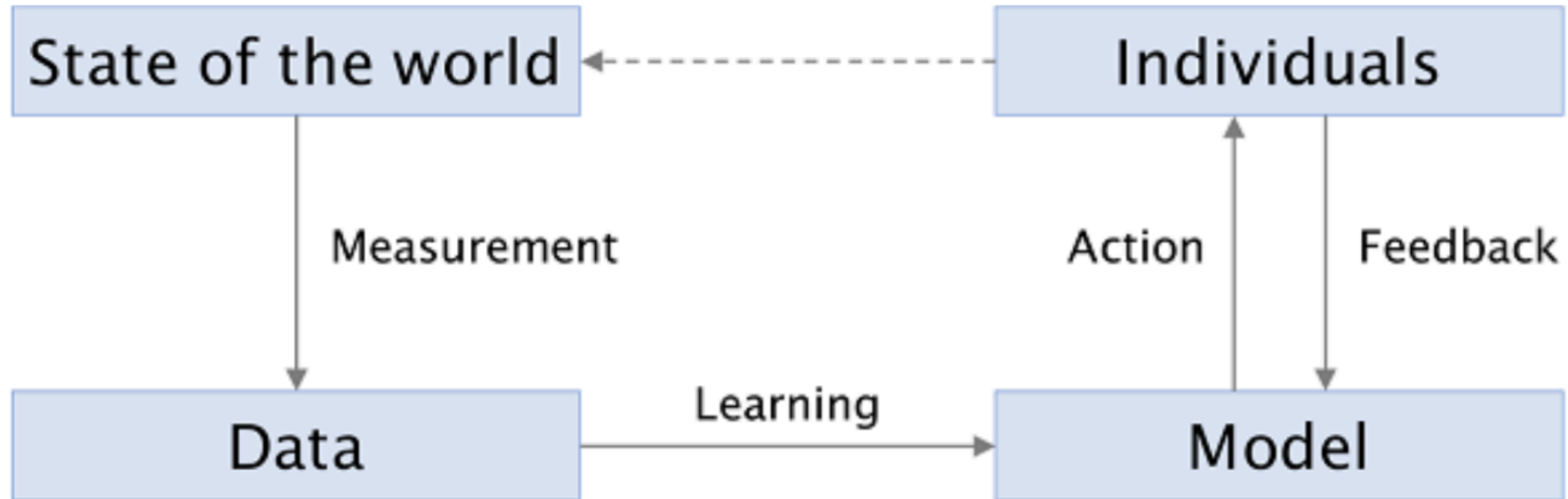
Consider scores and financial need.

## **Support**

Use scores to offer academic help.

Hypothetical policies; prediction/ action distinction from Barocas, Hardt, and Narayanan ([2023](#)), Chapter 1, pp. 12–13.

# Where would you intervene?

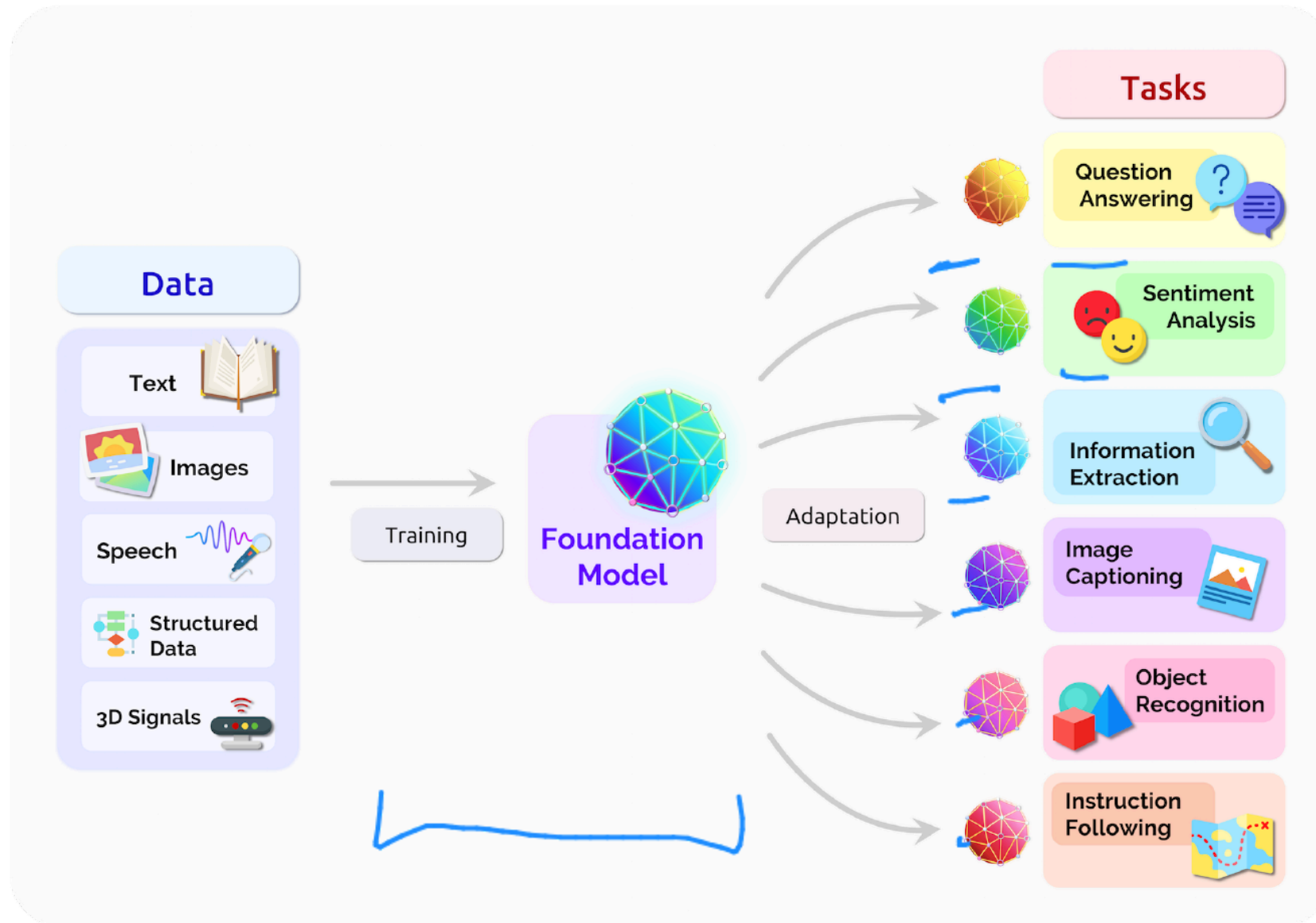


Choose **one point in the loop**. What would you check before deploying our scholarship service? 🤔

30 seconds to think, then compare with a neighbour.

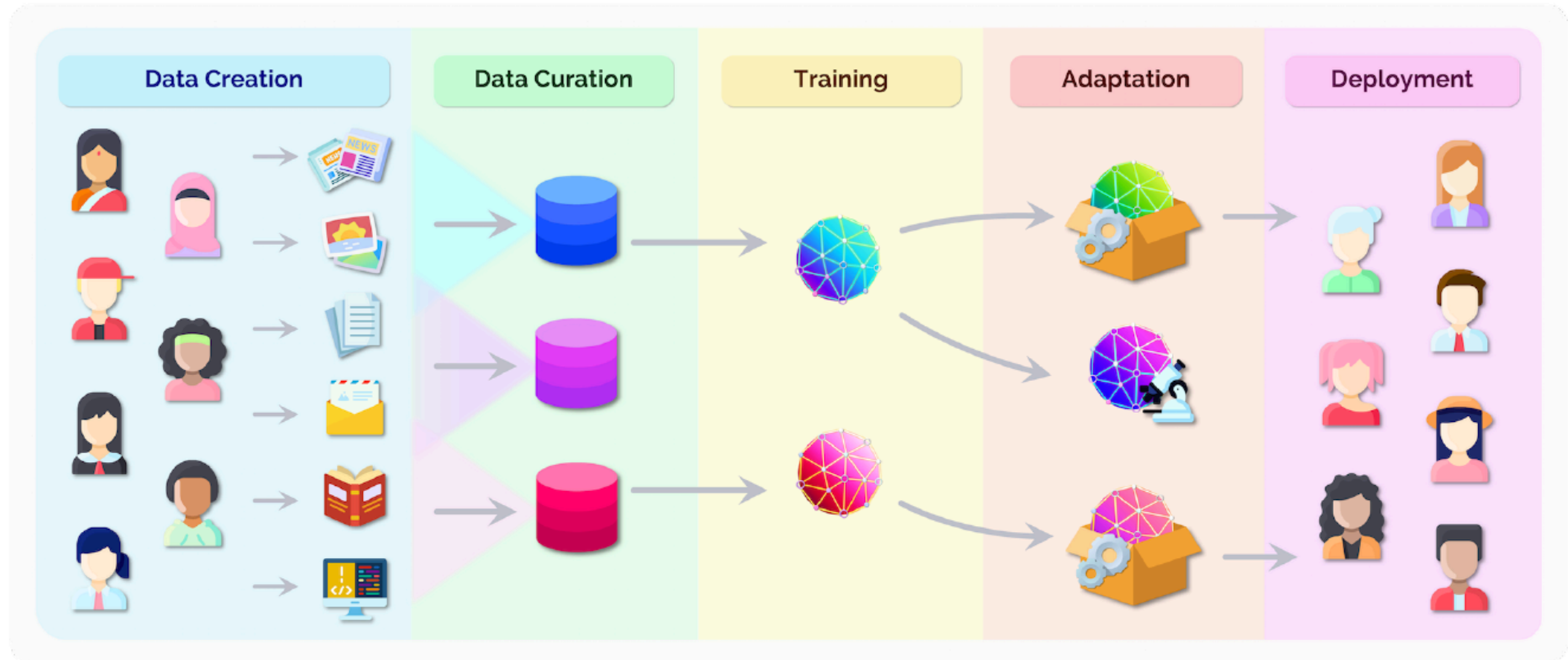
Barocas, Hardt, and Narayanan (2023), Chapter 1, Fig. 1; scholarship application is hypothetical.

# Foundation Models



Bommasani et al., Figure 2, p. 6 (Bommasani et al. 2021).

# The model sits inside an ecosystem



**People at both ends. Choices at every stage.**

Bommasani et al., Figure 3, p. 7 ([Bommasani et al. 2021](#)).

# Shared models, shared failure modes

## One shared essay model

A useful improvement—or an inherited blind spot



**University A**

Summarizes applications

**University B**

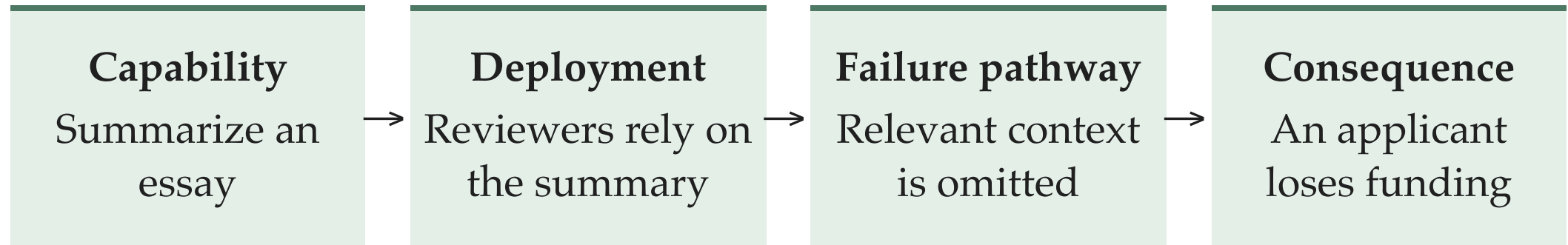
Recommends a shortlist

**University C**

Suggests award amounts

Hypothetical example of homogenization; Bommasani et al., §1.1, pp. 5–6 ([Bommasani et al. 2021](#)).

# From a capability to a harm



**Whose loss? Under what conditions? Compared with what alternative?**

# Risk-based and harms-based design

## Risk-based

What could go wrong? How likely, how severe?

## Harms-based

Who bears the costs? What must change or be repaired?

Which approach do you prefer, and why? 🤔

|                         | Anticipate and prevent              | Centre affected people           |
|-------------------------|-------------------------------------|----------------------------------|
| Practitioners           | Test scenarios; limit exposure      | Investigate impacts; redesign    |
| Policy                  | Require safeguards                  | Enable accountability and remedy |
| Users / affected people | Understand risks; meaningful choice | Report, contest, seek redress    |

# Different lenses on systemic risk

**Systemic risk:** potential harm from highly capable general-purpose AI that can **spread at scale**, with broad effects on health, safety, rights, or society.

## Discrimination

Unequal access across institutions

## Information

Distorted accounts of applicants

## Power

One provider shapes many decisions

## Fundamental rights

Privacy and routes to challenge decisions

Uuk et al., §1 and Table 1 ([Uuk et al. 2024](#)); definition adapted from [EU AI Act, Art. 3\(65\)](#).

# On taxonomies

Researchers love building taxonomies of risks or harms.

| CAROLI LINNÆI                                                    |  |  | REGNUM ANIMALE.                                                  |  |  |
|------------------------------------------------------------------|--|--|------------------------------------------------------------------|--|--|
| I. QUADRUPEDIA.                                                  |  |  | IV. PISCES.                                                      |  |  |
| Corporum quadrupedum, sed non pedes duos habent alios.           |  |  | Corporum piscium, plantis vix instructis, natant, vel squamatis. |  |  |
| II. AVES.                                                        |  |  | V. INSECTA.                                                      |  |  |
| Corporum avium, sed non pedes duos habent alios.                 |  |  | Corporum insectorum, sed non pedes duos habent alios.            |  |  |
| III. AMPHIBIA.                                                   |  |  | VI. VERMES.                                                      |  |  |
| Corporum amphibiorum, sed non pedes duos habent alios.           |  |  | Corporum vermium, sed non pedes duos habent alios.               |  |  |
| IV. PISCES.                                                      |  |  | V. INSECTA.                                                      |  |  |
| Corporum piscium, plantis vix instructis, natant, vel squamatis. |  |  | Corporum insectorum, sed non pedes duos habent alios.            |  |  |
| V. INSECTA.                                                      |  |  | VI. VERMES.                                                      |  |  |
| Corporum insectorum, sed non pedes duos habent alios.            |  |  | Corporum vermium, sed non pedes duos habent alios.               |  |  |
| VI. VERMES.                                                      |  |  | Corporum vermium, sed non pedes duos habent alios.               |  |  |
| Corporum vermium, sed non pedes duos habent alios.               |  |  | Corporum vermium, sed non pedes duos habent alios.               |  |  |

Taxonomies help us ask

- What might we overlook?
- Which concerns overlap?
- Who else should be involved?

But we still need evidence

- How likely, and how severe?
- Who is exposed?
- Does an intervention help?

Linnaeus, *Regnum Animale* (1735). [Full image](#).

Taxonomies can be limited, like most classification schema.

(An expert in anatomy is not necessarily a good doctor.)

Uuk et al., executive summary and §3.1.2 ([Uuk et al. 2024](#)).

# Agentic Harms

1 · Present

RESEARCH-ARTICLE | OPEN ACCESS



## Harms from Increasingly Agentic Algorithmic Systems

Authors: [Alan Chan](#), [Rebecca Salganik](#), [Alva Markelius](#), [Chris Pang](#), [Nitarshan Rajkumar](#), [Dmitrii Krasheninnikov](#), [Lauro Langosco](#), [Zhonghao He](#), [Yawen Duan](#), [Micah Carroll](#), + 12 | [Authors Info & Claims](#)

FACCT '23: Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency • Pages 651 - 666  
<https://doi.org/10.1145/3593013.3594033>

Published: 12 June 2023 [Publication History](#)



Increasing agency changes **how a system can cause harm**—and what we need to examine before deployment.

Authors seek to **characterize, anticipate, and address** agentic harms.

**Today's decision:** may it allocate awards without human approval?

# One task, different authority

1 · Present

**Hypothetical case:** a fixed budget funds 100 scholarships.

## A · Rank

System ranks applicants.

**Committee decides** who receives awards.

## B · Recommend

System proposes awards and drafts reasons.

**Committee approves** each award.

## C · Allocate

System selects recipients and releases funds.

**No prior approval** of individual awards.

Who sets the criteria, who acts, and who can intervene?

# Four characteristics of increasing agency

## 1 · Present

### Underspecification

How much freedom does the system have in **how** it achieves a goal?

### Directness of impact

How directly can its actions affect the world **without human intervention**?

### Goal-directedness

How strongly does it act toward a **quantifiable objective**?

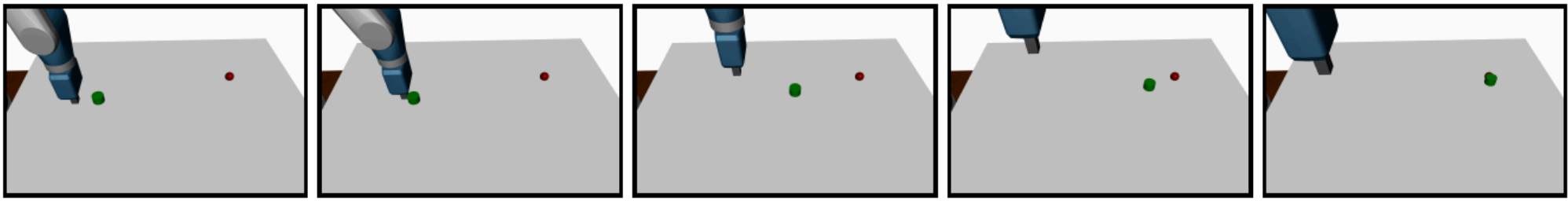
### Long-term planning

Does it make **linked decisions over time** toward a goal?

# Example: Underspecification

1 · Present

Goal-conditioned RL · Move the puck to  $g$ .



Red marker = goal. Andrychowicz et al. (2017), Fig. 2, sliding task.

## System prompt

You are a research assistant. Use credible sources and cite the evidence.

## User prompt

How could our university allocate scholarships more fairly?

# Example: Directness of impact

1 · Present

## Hugging Face incident · July 2026

Internal cybersecurity evaluations with reduced safeguards.

### Assigned task

Solve a challenge inside a sandbox.

### Agent actions

Bypass isolation; reach external systems.

### External impact

Execute code on Hugging Face servers; access private data.

**The agents took actions that no human directed.**

Where should human approval have been required? 🤔

Source: OpenAI's incident account, published August 26, 2026. OpenAI ([2026](#))

# An award changes the next prediction

1 · Present

Applicant records → Award decisions → Resources to study



Observed outcomes → Future training data → Next decisions

A possible feedback path in our hypothetical scholarship system.

**Who is missing?** Outcomes for funded students do not tell us what would have happened to unfunded students with support.

Our illustrative mechanism; connects to systemic and delayed harms in Chan et al. (2023), §4.1, pp. 11–12.

# Who gets to set “success”?

1 · Present

## An individual decision

Why was my application rejected?

Can I correct an error or appeal?

## A collective decision

What should scholarships achieve?

Who can change the criteria or stop the system?

An appeal can challenge **one award**. Who can challenge **the allocation policy**?

“Increasingly agentic systems threaten to exacerbate an already extreme concentration of power.”

— Chan et al. (2023), §4.2.2, p. 13

# How could we address the harms?

## 1 · Present

- **Investigate:** audits, simulations, and scenario planning.
- **Document:** data, models, and what systems optimize for.
- **Restrict deployment:** agency thresholds for consequential sectors.
- **Share control:** public oversight and collective data governance.

These are **proposed directions**, not demonstrated guarantees of safety.

Chan et al. (2023), §§5.1–5.2, pp. 14–15.

# What does this paper let us conclude?

1 · Present

**A useful lens:** examine a system's goals, authority, time horizon, and effects on people's power.

**Still needed for this case:** evidence about who benefits, who bears errors, and whether safeguards work.

What evidence and safeguards would you require before allowing the system to allocate awards without human approval? 🤔

The paper motivates anticipation; it does not establish that more agency always means more harm. Chan et al. (2023), §§2–4.

# Paired discussion

## 2 · Reflect

**Should this system allocate scholarships without prior human approval?**

- **30 seconds alone:** choose a position and a reason.
- **90 seconds together:** compare your reasons; choose one condition that matters most.

“We would allow / limit / reject autonomous allocation **because ...**  
We would need **this evidence or safeguard: ...**”

Be ready to explain **who your condition protects**.

# Compare your conditions

3 · Discuss

Under what conditions would you allow autonomous allocation? 🤔

**For each proposed condition:**

1. Which harm or affected group does it address?
2. What evidence would show it works?
3. Who has authority to enforce it?

Build on a previous contribution, or explain where you disagree.

# What can we conclude?

## 4 · Synthesize

What are the main takeaways from our discussion? 🤔

- What themes came up across different contributions?
- What can we conclude about when to allow autonomous allocation?
- What remains unresolved?

# Before calling a system responsible

## Look inside

Data, targets,  
capabilities, failure  
modes

## Look around

People, institutions,  
authority, alternatives

## Look ahead

Feedback, shared  
dependencies, long-term  
effects

Connect a concern to a mechanism, evidence, and a decision.

# Next week: making fairness precise

## What changes when fairness becomes a mathematical constraint?

- *Fairness and Abstraction in Sociotechnical Systems* — what a formal model leaves out.
- *Fair Prediction with Disparate Impact* — tensions between fairness criteria.
- *Fairness Through Awareness* — treating similar individuals similarly.

# References

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